

UPI PAYMENTS AND DIGITAL WALLET SYSTEM USING INTERFACE AND EXCEPTION HANDLING

PROGRAM

// UPI Payment and Digital Wallet System

// 1. PaymentService Interface

interface PaymentService

{

void pay(String receiverUPI, double amount) throws
InvalidUPIException,

InvalidAmountException, InsufficientBalanceException;

void checkBalance();

}

// 2. Custom Exception - Invalid UPI

class InvalidUPIException extends Exception

{

public InvalidUPIException(String message)

{

super(message);

```
}  
}
```

// 3. Custom Exception - Invalid Amount

```
class InvalidAmountException extends Exception  
{  
    public InvalidAmountException(String message)  
    {  
        super(message);  
    }  
}
```

// 4. Custom Exception - Insufficient Balance

```
class InsufficientBalanceException extends Exception  
{  
    public InsufficientBalanceException(String message)  
    {  
        super(message);  
    }  
}
```

// 5. Wallet Class

class Wallet

{

 // Private data members demonstrate encapsulation

 private String userName;

 private String mobileNumber;

 private String upild;

 private double balance;

 // Constructor

 public Wallet(String userName, String mobileNumber,
 String upild, double balance)

 {

 this.userName = userName;

 this.mobileNumber = mobileNumber;

 this.upild = upild;

 this.balance = balance;

 }

 // Add money to wallet

 public void addMoney(double amount)

 {

 if (amount > 0)

 {

```
        balance = balance + amount;

        System.out.println("Rs." + amount + " added successfully.");
    }
    else
    {
        System.out.println("Invalid amount!");
    }
}
```

```
// Get balance
public double getBalance()
{
    return balance;
}
```

```
// Get UPI ID
public String getUpild()
{
    return upild;
}
```

```
// Deduct money from wallet
public void deductMoney(double amount)
{
```

```

        balance = balance - amount;
    }

    // Display wallet details
    public void displayWalletDetails()
    {
        System.out.println("\n----- Wallet Details -----");
        System.out.println("User Name    : " + userName);
        System.out.println("Mobile Number : " + mobileNumber);
        System.out.println("UPI ID      : " + upid);
        System.out.println("Balance     : Rs." + balance);
    }
}

```

// 6. UPIpayment Class implements PaymentService

class UPIpayment implements PaymentService

```

{
    private Wallet wallet;

    // Constructor to associate Wallet with UPIpayment
    public UPIpayment(Wallet wallet)
    {
        this.wallet = wallet;
    }
}

```

```
}
```

```
// Pay method
```

```
@Override
```

```
public void pay(String receiverUPI, double amount)
```

```
    throws InvalidUPIException,
```

```
        InvalidAmountException,
```

```
        InsufficientBalanceException
```

```
{
```

```
    // Validate UPI ID using String operations
```

```
    if (receiverUPI == null ||
```

```
        !receiverUPI.contains("@") ||
```

```
        receiverUPI.startsWith("@") ||
```

```
        receiverUPI.endsWith("@"))
```

```
{
```

```
    throw new InvalidUPIException(
```

```
        "Invalid UPI ID! Payment cancelled.");
```

```
}
```

```
    // Validate amount
```

```
    if (amount <= 0)
```

```
{
```

```
    throw new InvalidAmountException(
```

```
        "Invalid amount! Amount must be greater than zero.");
```

```

    }

    // Check sufficient balance
    if (amount > wallet.getBalance())
    {
        throw new InsufficientBalanceException(
            "Insufficient balance! Payment cannot be processed.");
    }

    // Deduct payment amount
    wallet.deductMoney(amount);

    System.out.println("\nPayment Successful!");
    System.out.println("Receiver UPI : " + receiverUPI);
    System.out.println("Amount Paid : Rs." + amount);
}

// Check balance
@Override
public void checkBalance()
{
    System.out.println("Available Balance : Rs."
        + wallet.getBalance());
}

```

```
}
```

```
// 7. Main Class
```

```
public class Main
```

```
{
```

```
    public static void main(String[] args)
```

```
    {
```

```
        // Create a wallet
```

```
        Wallet userWallet = new Wallet(
```

```
            "Gayatri",
```

```
            "9876543210",
```

```
            "gayatri@upi",
```

```
            5000.00
```

```
        );
```

```
        // Create UPIpayment object and associate wallet
```

```
        UPIpayment payment = new UPIpayment(userWallet);
```

```
        // Add money
```

```
        userWallet.addMoney(2000);
```

```
        // Display initial wallet details
```

```
        userWallet.displayWalletDetails();
```



```
// Check balance
payment.checkBalance();

// Payment processing using try-catch-finally
try
{
    // Make UPI payment
    payment.pay("friend@upi", 1500);
}
catch (InvalidUPIException e)
{
    System.out.println("Error: " + e.getMessage());
}
catch (InvalidAmountException e)
{
    System.out.println("Error: " + e.getMessage());
}
catch (InsufficientBalanceException e)
{
    System.out.println("Error: " + e.getMessage());
}
finally
{

```

```
        System.out.println("\nTransaction process completed.");
    }

    // Display final wallet details
    userWallet.displayWalletDetails();

    // Display final balance
    payment.checkBalance();
}
}
```

OUTPUT

```
Amount Paid   : Rs.1500.0

Transaction process completed.

----- Wallet Details -----
User Name      : Gayatri
Mobile Number  : 9876543210
UPI ID         : gayatri@upi
Balance        : Rs.5500.0
Available Balance : Rs.5500.0
PS C:\Users\GAYATRI\eclipse-workspace\Java\src\javacode\java>
```